

China Dependence

India's battery storage journey is heavily dependent on Chinese manufacturers. The PLI Scheme announced by the government to promote local manufacturing is still nascent, with gaps in supply chain readiness, advanced technology adoption, and proven performance at scale.

Sinha says that while government initiatives exist to promote local manufacturing, the current policy emphasis is skewed toward electric mobility rather than BESS.

"Importing high-quality cells and battery management systems from established global players—while localising the assembly of racks (battery), energy management systems, and containerisation—offers a balanced solution that ensures performance reliability, cost optimisation and reduced exposure to Customs duties," says Sinha. He adds that as India's BESS market scales up, investments in advanced technologies, robust supply chains, and recycling infrastructure will be essential to reduce geopolitical risks, enhance energy security, and align with national priorities. The government has announced a PLI programme to support the manufacturing of 50GWh Advanced Chemistry Cell batteries with a budgeted outlay of ₹18,100 crore.

Risk Analysis

IESA's Debmalya points out that India does not have a long history of battery storage. "Just 0.5GWh is deployed. Another 1.2GWh is about to come out by the end of the year. How these batteries perform in the operational phase will need to be seen. Many of these players have worked on very competitive



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Aggressive underbidding not only skews the auction but also places the successful commissioning of projects in jeopardy**

PRAVEER SINHA
Chairman & MD, Tata Power



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DEBMALYA SEN
President, India Energy Storage Alliance

numbers. This happened with solar also. With all those low tariffs, half never got realised and the projects never came up,” he says.

The capacity expansion target and tariffs are encouraging, but how much actually get operational and sustained will be key in the coming years. “These are all Chinese cells. They give you a warranty on standard temperature and pressure. You will be installing those in a place where the average temperature is around 50 degrees, like Rajasthan and Gujarat, or maybe in the moist climate of Kerala and Tamil Nadu,” says Debmalya.

“Every assumption, everything that you have thought about, is on paper. How will they perform on the ground? No one knows,” he added. In this context, the long-term financial viability of BESS projects assumes significance. “The dominance of lithium-ion technology also poses lifecycle limitations, as most renewable PPAs in India are structured for 25-year horizons, whereas current battery technologies require major replacements within that timeframe—raising concerns around long-term viability and financial structuring,” warns Sinha.

To achieve India's ambitious target of 400 GWh of energy storage by 2031-32, industry participants say policy must strike a balance between promoting domestic manufacturing through initiatives such as PLIs and enabling access to the most advanced global storage technologies. **BT**

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